

RND-6: SAFE IN-TIME COVARIATES ??? CUPAC-STYLE ????????

> ??????: ? ??????. ????????????? ?????????????? ????????? ? ??????? ???????
> A/B ? CUPAC A ? A+B ? A+B+C; ?????????? ?????????? ?????????? open-source ?????????? ?
> ?????????? ?????????????? ?????????? ? delta-???????????? ?? A-only (??????? «Expanded public dataset
> validation»). ?????????? ??????? ?? B/C-???????? ? notebook.ipynb (???????????, seed=11).

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???????????, ??? ? ? ?????????????????? ??????????? in-time ?????????? ??????
scikit-CUPAC ?????????????????? ?????????? ?????????? (? ??????????? ??????????) ???
?????????? ?????????????????? ?????????????? ??????? ATE.

?????? ?????????? (SAFETY LAYER)

? **A** pre-treatment ? ??? ? CUPAC;
? **B** expert-safe in-time ? ?????????? second-stage ??????????;
? **C** balance-gated in-time ? ?????????? ?????????? ????? balance/missingness gate;
? **D** dag-required ? ??? estimator (????? causal-????????);
? **E** mediator-risk, **F** leakage ? ?????????? ??? ?????????? ?????? ATE;
? **unsafe_demo** ? ?????? ?????????????? ??????, ?? ??????????

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ab_hypex (A/B baseline) • hypex_cupac (reference/parity, ??? ??????) •
sklearn_cupac_A ? sklearn_cupac_A_plus_B_linear ? sklearn_cupac_A_plus_B_plus_C_linear
(????????? ?????????? ??????????) • unsafe_demo_optional (?????????????).

?????????? ? ? ?????????? (N=8000, SEED=11, ?????????? ATE = 2.8)

method	ate	p_value	variance_reduction_vs_ab_%	incremental_vs_predecessor_%	safety_status
ab_hypex	2.758	0.0	0.0	?	ok
hypex_cupac	2.785	0.0	13.56	?	reference_only
sklearn_cupac_A	2.801	0.0	25.63	25.63	ok
sklearn_cupac_A_plus_B_linear	2.775	0.0	42.38	22.52	ok
sklearn_cupac_A_plus_B_plus_C_linear	2.766	0.0	52.80	18.07	ok
unsafe_demo_optional	1.000	0.0	83.17	?	unsafe_demo

(????????? ?????????? ????????? ? ?????? ?????????????? ?????????? ?? ?????????? ??????????)

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- ?????? ?????????????? ??????? ???? ?????????????????? ?????????: A (~26%) ? +B (~42%) ? +B+C (~53%)
????????? ?????????? ?????????????? A/B, ??? ??? ATE ?????????? ?????????????? (? 2.8).
- CUPAC parity: hypex_cupac (~14%) ?????????? ?? ?????? ? sklearn_cupac_A;
????????? ?????????? ??????? ??? R&D ? ?????????????? ??? ??????????
- Balance/missingness gate ?????????? A/B/C (|SMD|>0) ? ?????????? D/E/F ? unsafe_demo
(|SMD| ? 0.45?1.14, p>0) ? ?????????, ?????? ?? ?????? ?????? ? estimator.

4. unsafe_demo ???? ???????????? «????????? ??????????» (~83%), ?? ??????? ?????? ATE
?????? ?? ?????????? 2.8 ? ?????????? ? 1.0 (??, ????????) ? ?????????????? ???????
outcome-derived ??????????; ? ?????????????? ??????? ?? ???????.

?????????????? ? ??????????????

? balance-test ? ?????????????? gate (????? ?????????????? ?????????? E/F), ? **??** ?????????????????
?????????? ??????????????; ??? D-?????????? ?????? ????????? DAG/causal-?????
? E/F ??????????; safe-intime ?????????????? **??????** ? ?????????? second-stage, ?? ?
???????????????? ML-????????.

??????? ??????????? ?? ?????????? ???????

? **??????????** ?????????????? ??????? ????????? **A+B+C** (????????? ?????): ?????? ??? ??????
A/B/C/D/E/F ?????????? ?? ??????????????
? **??????????? ?????????? ?????????? ??????? ?????????????? ? ?????????? A-only** CUPAC-?????????????
????? ?????? (download + schema_summary) ?????? Orange Belgium, Lenta, Criteo, X5
RetailHero,
MegaFon, Lazada/DESCN, CUVET-policy, MT-LIFT, Dunnhumby ? semi-synthetic causal-?????????????
? **Hillstrom** ?????????? pipeline/A-only, **??** B/C.
? ?????????? safe in-time **B/C** ??????????????: ?????????? ??? ? ?????????? ('f*', 'PC*', 'X_*'),
?????
??? ???-????????? ?????????? (A), ??? ????-treatment ?????????? (E/F). ?????????? ? ??????????????
????????????????? ??? ? ?????????? ??? ? ?????????? (X5), ??? ? ? ?????????????????? (Dunnhumby).
? **????????? A+B+C-????????????? ? ? ?????????? ?????????? ?????????? ? ? ?????????????????** ? ??????????
????????????????????, ??? ? ?????? ?????? ? ? ?????? ?????????? B/C-????????? ? timing-evidence.
? ?????? ? ?????????????? ?????????????????? ??????????: **docs/06_real_dataset_raw_audit.md**.

EXPANDED PUBLIC DATASET VALIDATION

?????????????? ?????????? (????? ??? ? ??????????????? uplift-????????? + ?????????? ??????????/????????????? +
????????????? ?????????????? + ?????????????? delta-????????? ?????????). ??????? ?????? ? ?????????
?????????????????????????: docs/06_expanded_dataset_scouting.md.

????????????????????????? ?????????? ?????????? (?? ?????? ????, ?? ?? ?????????????????):
? **A** (pre-treatment) ? ?????????????? ??? ??????????: **9 ?????????** ?????????? (+ ??????????).
? **B** (expert-safe in-time) ? **0**?: ?? ??? ? ?????????? ?????? ?? ?????????????? ??????????????,
????????????????? ????????????????????? in-time ??????????-????????????? ??????????
? **C-????????????** (?????????, ??????????) ? **0**; sandbox-????????????? C ?????????????? ??????????
?? 2 event-log ?????????? (Open Bandit, dunnhumby), ?? ??? ? ? ?????????? ??????????????
? **D-????????????? ? **5** (Criteo 'exposure'; Lazada biased-train propensity; Open Bandit
action/position/pscore; CriteoPrivateAd display/privacy-?????????; dunnhumby
campaign/coupon).
? **E/F/unsafe-????????????** ? **7+** (Hillstrom 'visit'/'conversion'; Criteo 'exposure';
X5 ???-????????????????????? ?????????; Open Bandit reward-????????????????; CriteoPrivateAd delayed
click/sale; dunnhumby redemption ? ??).

???????-A ?????????? (real RCT/uplift): Hillstrom, Lenta, X5 RetailHero, Orange Belgium,
Lazada/DESCN (???????????????????????? test), Criteo (??????? A-only).
C/D-«????????????» (?? A/B): Open Bandit (logged bandit), dunnhumby Complete Journey
(observational), CriteoPrivateAd (advertising logs) ? ??????? ?? ? *?????????????????
C/D-?????????,

?? ?????????? causal-????????????? safe in-time.

> ?????? A+B+C-????????? ?? ?????????? ?????????? ?????????? ?? ??????????????. ??????
> C/D-????????????????????? ?????? ?????????????? ?? event-log / bandit / advertising / journey
> datasets, ???????? ?? ?????????? ???????? A/B uplift benchmark.

Delta-????????? ?? ?????????? ???????? (A-only ?????????? ab_hypex ? sklearn_cupac_A, seed=11).
????????? ? ???????? ?????????????? python tools/audit_datasets.py --delta; ?????????????? ?????????????? ?
expanded_dataset_delta_results.csv, ?????????? ? figures/.

dataset	validation_level	n	ATE (A/B)	var.reduction	CUPAC-A vs A/B, %	A-?????????????	safety
hillstrom	A_only_real	64 000	0.597	0.06	18	ok	
orange_belgium	A_only_real	11 896	?0.0028	6.06	336	ok	
lazada_descn	A_only_real	181 669	0.0037	5.41	83	ok	
lenta	A_only_real	687 029	0.0075	14.36	196	ok	
x5_retailhero	A_only_real	200 039	0.0332	16.10	10	ok	

? **CUPAC-A ???? ?????????? ?????????? ?????????? 0?16%** ?? ?????????? ????????; ?????????? ?????? ???,
??? ?????? ?????????????????? pre-treatment ?????????? ? lenta (??????????????? ???-????????? ??????????,
~14%) ? X5 (engineered ?????????? ??????????, ~16%). ATE ?????????? ?????????????? (?????????????????).
? **hillstrom** ? pipeline/A-only smoke: ?????????? 'spend' ?????????? ???????? A-??????? ??????
????????? (~0.06%).
? **??????????? '+B' ? '+B+C' ?? ?????????? ?????????? ?? ?????????????????**?: ?????????? B/C-????????? ???,
?????????
????????????????? delta-????????? B/C ??-????????? ?????????????????? ?????????? ?? ?????????????? (?????????
?????).
? **?? ?????? ? delta-?????????:** 'criteo' ? ?????????? ?????????? HTTP 403 (access_blocked);
open_bandit ? ?????? obp ?????????????????? ? pandas 2.x (DataFrame.drop("col", 1)),
runtime-blocked sandbox; dunnhumby / criteo_private_ad ? ?????????? ?????????????? ??????????????????
registration-gated ?????????? ? ??? safe B/C ?? ?????????????????? (????????? ?????????? ??????????
?????????).

????????????????? ?? ?????????? ? HYPEX

????????? ?????????????????? ?????????????? ?????????? ?? ?????????????? ? ?????????????? ? HypEx CUPAC. ??????
?????????????
? HypEx ??????????????: (1) ?????????????????? ?????? B/C ?? ?????????????????? ?????????? open-source ??????????????
(2) ?????????????????????? balance/missingness gate ??? ?????????????????????? ??????????????; (3) ??????????????????????
????????? E/F ? ?????????????????? unsafe_demo. ?????????????????? ?????????????????????? CUPAC ??
????????????? ?????????? ? VarWar (????????????? ?????????????? ~34% ? ~33%; ?????????? ??????????????, ?? ??????????????????).

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bash
pip install -e .[ml]
??. NOTEBOOK.IPYNB; ?????????? ATE-CI ? ?????????? ?????????????????? ?? RND_REPORTS.BENCHMARK

DELTA-????????????? ?? ?????????????? ?????????????? (????????? GITIGNORED DATA/, RAW ?? ??????????????????):

python tools/audit_datasets.py --delta # ? expanded_dataset_delta_results.csv + figures/

python tools/generate_pdf.py

??????????? report.pdf